

MAC PHAM THIEN LONG

AI ENGINEER

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PROFESSIONAL SUMMARY

A Management Information Systems student with a technical foundation in Multi-agent system orchestration, LLM optimization, and data engineering. Specialized at developing secure, production-ready AI applications—ranging from localized coding assistants to large-scale intelligent tutoring systems—with a strong focus on system robustness, safety guardrails, and data privacy. Driven to leverage skills in statistical analysis, MLOps, and trustworthy AI to research and deploy reliable, enterprise-grade intelligent systems.

TECHNICAL SKILLS

- **Programming Languages:** Python (Pandas, NumPy, SciPy), SQL (PostgreSQL, MySQL), C++.
- **AI / ML / NLP:** PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, NLP, RAG, Basic Computer Vision. LLM Fine-tuning (**QLoRA**), AI Agents & Function Calling.
- **MLOps & Data Engineering:** Docker, Docker Compose, Kubernetes, Apache Kafka, Apache Spark, Feast Feature Store, MLflow, Vector Databases (Qdrant), TimescaleDB.
- **Domain Expertise:** Data Cleaning, Feature Engineering, Quantitative Analysis, Time-series Forecasting, Simulation & Backtesting.
- **Soft Skills:** Logical & Critical Thinking, Independent & Teamwork, Fast Learner of Emerging Technologies.

EDUCATION

International School - Vietnam National University, Hanoi

Bachelor of Management Information Systems

• Cumulative GPA: **3.70 / 4.0**

2022 – Present

Expected graduation: 8/2026

RESEARCH & WORK EXPERIENCE

VinSmartFuture - Device AI Division

04/2026 – 07/2026

AI Engineer Intern (Conversational AI & NLP Department)

Hanoi, Vietnam

- **Co-developed** a LangGraph-based in-car voice assistant using a registry-validator-executor architecture (MRKL-based) and Llama Guard 3 on NVIDIA NIM for input/output moderation.
- **Conducted adversarial testing (red-teaming)** on tool calls (using ToolSword framework taxonomy) to evaluate agent resilience against prompt injections, risky tool cues, and malicious tool executions.
- **Designed and implemented** the [vivi-brain-bench](#) testing framework to evaluate 20 control tools across 90 tasks (270 trials) under strict **Pass³ consistency** metrics, achieving a **95.56% tool execution success rate**.
- **Achieved a 47.78% Pass³ rate** overall (**77.78%** on base tasks), while ensuring safety compliance via Human-in-the-Loop (HITL) gates and achieving **100% classification accuracy** on safety guardrail trials.

Viettel Networks

11/2025 – 12/2025

AI Engineer Intern (Viettel AI Fresher Program 2025)

Hanoi, Vietnam

- **Developed** a secure, offline (local) AI coding assistant for edge computing, ensuring zero data leakage and maintaining user data privacy in safety-sensitive, air-gapped environments.
- **Fine-tuned** a lightweight Qwen2.5-Coder model using **QLoRA** and **Direct Preference Optimization (DPO)**, boosting code completion accuracy to a **40%–50% Exact Match (EM) rate** and **75%–85% Edit Similarity** on the test set.
- **Built a data sanitization pipeline** to scrub secrets (API keys, IPs) and PII, transforming raw code from "The Stack" dataset into Fill-in-the-Middle (FIM) format.
- **Optimized** edge deployment using 4-bit **GGUF quantization** (Q4_K_M) to fit a **373.7MB** memory footprint, delivering a generation throughput of **23.87 tokens/sec** on CPU (20–50ms latency).

Worldquant BRAIN Vietnam

Part-time Research Consultant

07/2023 – Present

Hanoi, Vietnam

- **Designed and optimized over 667 alpha models** (predictive financial signals) using statistical arbitrage and machine learning to forecast market movements.
- **Processed and analyzed** large-scale financial time-series datasets to identify market inefficiencies using Python (Pandas, NumPy) for simulation and backtesting.

SELECTED PROJECTS

Graduation Thesis: Multi-Agent Financial Analysis System

01/2026 – 07/2026

AI Research & Development

- **Orchestrated** a multi-agent LangGraph workflow featuring a query router, a ChromaDB retriever, and a sandboxed python execution environment to parse and evaluate PDF financial reports.
- **Implemented** an asynchronous pipeline using FastAPI, PyMuPDF, and OCR (Pytesseract, EasyOCR) to extract tables and text from multi-page documents.
- **Evaluated** system robustness using RAGAS on a 200-sample benchmark, achieving over 93% Faithfulness and 95% Answer Relevance to mitigate hallucinations.
- **Technologies:** LangGraph, FastAPI, ChromaDB, Redis, Docker, React, TypeScript.
- **Links:** [Github](#) | [Demo](#)

EduPilot: Large-Scale AI Teaching Assistant System (9000+ Students)

03/2026 – 05/2026

AI Research & Development

- **Designed and scaled** a Socratic tutoring agentic system supporting **9,000+ students** (optimizing resource load from 1 TA per 1,800 students) and automating up to **80%** of repetitive academic queries.
- **Reduced response latency** from 2–3 days to **under 1.5 seconds** using advanced RAG and a Redis-based semantic cache that successfully intercepted **30%–50%** of duplicate queries.
- **Engineered dual-stage guardrails** (gpt-4o-mini) that achieved a **92%–95% intent classification accuracy** and **93% context retrieval precision** on a 100-sample ground-truth benchmark.
- **Implemented a data-privacy gateway** to redact PII (student IDs, contact details) and integrated a graceful fallback system for human TA escalation when confidence score dropped.
- **Technologies:** LangGraph, Python, OpenAI API, Redis Semantic Cache, pgvector, Pydantic Validator.
- **Links:** [Github](#)

Reefer Logistics Supply Chain Optimization Model

05/2025

Academic Research Paper

- **Formulated** an MIQP optimization model for Vietnam's domestic reefer logistics market, resolving vehicle routing, cold storage selection, and perishability loss constraints.
- **Linearized** binary-continuous variable interactions in model constraints to enable efficient solving via Gurobi.
- **Derived** an optimal network (1 farm, 1 warehouse, 5 trucks) that cut costs to 257.38M VND (saving 20–35% over decentralized baselines) and eliminated delivery delay penalties.
- **Technologies:** Python, Gurobi, Mixed-Integer Quadratic Programming, Supply Chain Simulation.

Fire Detection System

05/2024

Academic Research Paper

- **Designed and built** a hybrid safety-critical fire detection system combining hardware sensor arrays (temperature, smoke MQ-2, flame) with real-time computer vision on a Raspberry Pi.
- **Trained and evaluated** custom YOLOv5 models on a Roboflow dataset of 2,427 fire/smoke images, achieving **85.5% mAP@0.5** (56.8% mAP@0.5:0.95) with **89.5% precision** and **80.6% recall** at a threshold of 0.5 IoU.
- **Conducted comparative benchmarking** of model variants under low-compute edge constraints, demonstrating the performance of **YOLOv5x (90.5% mAP)** over **YOLOv5n (83.2% mAP)**.
- **Presented & Published** the paper at the 5th FBU Student Scientific Research Conference, demonstrating early warning feasibility in residential and industrial settings.
- **Technologies:** YOLOv5, Roboflow, Google Colab, Embedded Systems, Raspberry Pi 4.

HONORS AND AWARDS

- **12/2025:** "Student of 5 Merits" Award – Hanoi City Level.
- **09/2025:** Vietnam Representative, **International Financial Security Olympiad - Final stage**, Krasnoyarsk, Russia.
- **08/2025: 2nd Prize**, MIS Talent Competition - E-Mastery (AI applications in business models).
- **07/2025: 2nd Prize (National Finals)**, Microsoft Office Specialist World Championship (MOSWC 2025) – Microsoft Excel Category.
- **05/2025: 3rd Prize, Co-Author**, 17th Student Scientific Research Conference (VNU-IS) – Topic: "*Optimization model for the cold-chain logistics market in Vietnam*".
- **11/2024: 1st Prize**, Star Awards English Competition – Hanoi Region.
- **05/2024: Presented & Published, Co-Author**, 5th Student Scientific Research Conference (FBU) – Topic: "*Multi-Sensor Real-Time Fire and Smoke Detection System using YOLOv5*" (Selected for conference proceedings).

CERTIFICATIONS

- **07/2026:** Completed the **Vingroup Applied AI Talents Program, Cohort 1**
- **12/2025:** Completed the **Viettel AI Fresher Program 2025**.
- **08/2025:** TOEIC Certificate: **880 / 990**.
- **04/2025:** Microsoft Office Specialist: Excel 2019 (**1000/1000**).
- **03/2022:** English Proficiency: **VSTEP B2**